The Relationship Between Children's Mathematics Interest and Mathematics Skills

Final assignment
Basic course for structural equation modeling

1. Introduction

The purpose of this study / final assignment is to examine the relationships between children's interest in mathematics and their mathematical skills. The analysis focuses on identifying associations between the variables measured in the study and exploring whether selected variables can explain variations in other variables over time. The assignment was conducted primarily using Python, with additional analyses performed in Mplus to better understand and validate the CFA model.

The study uses three statistical approaches. First, descriptive statistics are used to provide an overview of the data, including key summary measures, missing data, covariances, and correlations between variables. Second, regression analysis is used to examine the relationship between children's mathematical interest in autumn and their mathematical skills in spring, while accounting for their mathematical skills in autumn. Third, confirmatory factor analysis (CFA) is used to evaluate whether three observed indicators of mathematical interest sufficiently represent an underlying latent construct at both measurement occasions (autumn and spring).

The research questions are:

1. Does children's interest in mathematics in autumn predict their mathematics skills in spring, after controlling for their mathematics skills in autumn?
2. Do three observed indicators of mathematics interest measure a latent mathematics interest construct in autumn and spring?

2. Descriptive Statistics

2.1 Dataset and Descriptive statistics

The first stage of the analysis examined the dataset using descriptive statistics. The original dataset consisted of 167 participants. However, the number of valid responses varied across variables due to missing data, with valid observations ranging from 159 to 165.

A total of nine variables were included in this study. These consisted of mathematics interest in autumn (miema), mathematics skills in autumn (matisum_1) and spring (matisum_2), as well as the six indicators measuring enjoyment of mathematics activities (mie1c and mie1c_2), willingness to participate in mathematics activities at school (mie2c and mie2c_2), and willingness to participate in mathematics activities at home (mie3c and mie3c_2) in autumn and spring.

Table 1 presents the descriptive statistics for the nine variables, including the number of valid observations, mean, standard deviation, minimum, maximum, median and quartiles. These statistics were calculated to examine the distributions and variability of the study variables.

The mean values ranged from 4.157 to 19.552, while standard deviations ranged from 0.862 to 5.033. The differences in means and standard deviations reflect the different measurement scales used for mathematics skills and mathematics interest indicators.

The median values ranged from 4.667 to 5 for all seven indicators measuring interest in mathematics, suggesting that participants generally reported relatively high levels of

mathematics interest. Also, mean and median for mathematics skills grew from autumn to spring, suggesting possible improvement on skills. However, this descriptive difference alone does not establish a statistically significant improvement in mathematics skills.

	N	Mean	SD	Min	Q1	Median	Q3	Max
miema	160	4,274	0,862	1	3,667	4,667	5	5
matisum_1	161	14,745	5,033	0	12	15	18	29
matisum_2	165	19,552	4,933	2	17	20	22	31
mie1c	160	4,35	1,089	1	4	5	5	5
mie2c	159	4,327	1,04	1	4	5	5	5
mie3c	159	4,157	1,145	1	4	5	5	5
mie1c_2	165	4,261	1,115	1	4	5	5	5
mie2c_2	165	4,345	1,063	1	4	5	5	5
mie3c_2	165	4,261	1,076	1	4	5	5	5

Table 1. Descriptive Statistics of the study variables (Python: pandas).

2.2 Missing data

Table 2 presents the missing data analysis, showing the number and percentage of missing values for each of the nine study variables. The results indicate that mie2c and mie3c had the highest amount of missing values, with 8 missing observations (4.79%) each. In contrast, matisum_2, mie1c_2, mie2c_2, and mie3c_2 had the lowest amount, with only 2 missing observations (1.20%) each. Overall, this suggests a higher response rate in spring than in autumn.

	Missing N	Missing %
miema	7	4,192
matisum_1	6	3,593
matisum_2	2	1,198
mie1c	7	4,192
mie2c	8	4,79
mie3c	8	4,79
mie1c_2	2	1,198
mie2c_2	2	1,198
mie3c_2	2	1,198

Table 2. Number and percentage of missing observations for each study variables (Python: pandas).

Table 3 presents the missing data patterns across the study variables. The results show that most participants had complete data across all variables, while only a small number of participants had between one and three missing values. Overall, the amount of missing data was relatively low, suggesting a high level of data completeness.

miema	matisum_1	matisum_2	mie1c	mie2c	mie3c	mie1c_2	mie2c_2	mie3c_2	Freq
0	0	0	0	0	0	0	0	0	156
1	1	0	1	1	1	0	0	0	5
1	0	0	1	1	1	0	0	0	2

0	0	1	0	0	0	1	1	1	2
0	1	0	0	1	0	0	0	0	1
0	0	0	0	0	1	0	0	0	1

Table 3. Missing data patterns across the study variables (Python: pandas).

2.3 Correlations, Covariances and Covariance coverage

Table 4 presents the Pearson correlation coefficients between the nine variables. The results show strong positive correlations (0.744-0.831) between miema (the mean score of mie1c, mie2c, and mie3c) and its component variables. This is expected because miema is calculated from these variables. Overall, the results suggest that participants who scored higher on one measure tended to score higher on related measures, indicating positive associations between the variables.

	miema	matisum_1	matisum_2	mie1c	mie2c	mie3c	mie1c_2	mie2c_2	mie3c_2
miema	1	0,139	0,203	0,797	0,831	0,744	0,349	0,363	0,274
matisum_1	0,139	1	0,748	0,109	0,158	0,067	0,232	0,143	0,091
matisum_2	0,203	0,748	1	0,12	0,23	0,143	0,265	0,21	0,103
mie1c	0,797	0,109	0,12	1	0,586	0,316	0,264	0,395	0,198
mie2c	0,831	0,158	0,23	0,586	1	0,41	0,239	0,265	0,159
mie3c	0,744	0,067	0,143	0,316	0,41	1	0,313	0,195	0,273
mie1c_2	0,349	0,232	0,265	0,264	0,239	0,313	1	0,803	0,731
mie2c_2	0,363	0,143	0,21	0,395	0,265	0,195	0,803	1	0,657
mie3c_2	0,274	0,091	0,103	0,198	0,159	0,273	0,731	0,657	1

Table 4. Pearson correlation matrix of the study variables (Python: pandas).

Table 5 presents the covariance matrix for the nine variables. All covariances are positive, suggesting once again that the variables generally move in the same direction. This means that higher values in one variable tend to be associated with higher values in another. The differences in covariance values between skills and interests reflect variations in the scales.

	miema	matisum_1	matisum_2	mie1c	mie2c	mie3c	mie1c_2	mie2c_2	mie3c_2
miema	0,743	0,587	0,836	0,747	0,746	0,732	0,339	0,335	0,259
matisum_1	0,587	25,328	18,717	0,582	0,801	0,374	1,295	0,767	0,48
matisum_2	0,836	18,717	24,334	0,625	1,113	0,786	1,459	1,101	0,544
mie1c	0,747	0,582	0,625	1,185	0,665	0,393	0,325	0,46	0,236
mie2c	0,746	0,801	1,113	0,665	1,082	0,484	0,269	0,285	0,171
mie3c	0,732	0,374	0,786	0,393	0,484	1,311	0,4	0,232	0,339
mie1c_2	0,339	1,295	1,459	0,325	0,269	0,4	1,243	0,952	0,877
mie2c_2	0,335	0,767	1,101	0,46	0,285	0,232	0,952	1,13	0,751
mie3c_2	0,259	0,48	0,544	0,236	0,171	0,339	0,877	0,751	1,157

Table 5. Covariance matrix of the study variables (Python: pandas).

Table 6 presents the covariance coverage, showing the amount of observations availabe for each pair of variables when calculating covariances. The values range from 0.940 to 0.988,

indicating that between 94.0% and 98.8% of observations were available. This suggests that missing data had relatively small impact on the available data for estimating the covariances.

	miema	matisum_1	matisum_2	mie1c	mie2c	mie3c	mie1c_2	mie2c_2	mie3c_2
miema	0,958	0,952	0,946	0,958	0,952	0,952	0,946	0,946	0,946
matisum_1	0,952	0,964	0,952	0,952	0,952	0,946	0,952	0,952	0,952
matisum_2	0,946	0,952	0,988	0,946	0,94	0,94	0,988	0,988	0,988
mie1c	0,958	0,952	0,946	0,958	0,952	0,952	0,946	0,946	0,946
mie2c	0,952	0,952	0,94	0,952	0,952	0,946	0,94	0,94	0,94
mie3c	0,952	0,946	0,94	0,952	0,946	0,952	0,94	0,94	0,94
mie1c_2	0,946	0,952	0,988	0,946	0,94	0,94	0,988	0,988	0,988
mie2c_2	0,946	0,952	0,988	0,946	0,94	0,94	0,988	0,988	0,988
mie3c_2	0,946	0,952	0,988	0,946	0,94	0,94	0,988	0,988	0,988

Table 6. Covariance coverage matrix for the study variables (Python: pandas).

3. Regression Analysis

3.1 Research question and method/model

The analysis was conducted in Python using multiple linear regression (ordinary least squares, OLS). The analysis included 157 observations. Cases with missing values in any of the three variables were excluded from the analysis. The regression model estimated the association of autumn mathematics skills and autumn mathematics interest with spring mathematics skills.

The research question was whether children's mathematics interest in autumn (miema) predicts their mathematics skills in spring (matisum_2), after controlling for mathematics skills in autumn (matisum_1).

Regression equation:

$$\text{matisum_2} = \beta_0 + \beta_1 \text{matisum_1} + \beta_2 \text{miema} + \epsilon$$

Here, matisum_2 represents mathematics skills in spring, matisum_1 represents mathematics skills in autumn, and miema represents mathematics interest in autumn. The regression coefficients (β_1, and β_2) show how a one-unit increase in each predictor is associated with a change in mathematics skills in spring, while keeping the other predictor constant. The intercept (β_0) represents the predicted spring mathematics skills when both predictors are zero.

3.2 Results

The regression model was statistically significant (F=89.86, p<0.001) and explained 53.9% of the differences in mathematics skills in spring (R^2 = 0.539). This means that mathematics skills in and interest in autumn together helped explain more than half of the differences in students' mathematics skills in spring.

Mathematics skills in autumn (matisum_1) were a statistically significant positive predictor of mathematics skills in spring (B=0.700, SE=0.054, t=12.879, p<0.001). This means that a one unit increase in mathematics skills in autumn was associated with a 0.700 unit increase in mathematics skills in spring, when mathematics interest in autumn was held constant.

Mathematics interest in autumn (miema) also had a positive coefficient (B=0.595, SE=0.304, t=1.957, p=0.052), but its association was not statistically significant at the 5% level. In addition, the confidence interval included zero ([-0.006, 1.197]), suggesting that the relationship between mathematics interest in autumn and mathematics skills in spring was not statistically significant. Therefore, there was insufficient evidence that mathematics interest in autumn independently predicted mathematics skills in spring when mathematics skills in autumn were held constant.

In summary, the regression model results suggest that mathematics skills in autumn were an important predictor of mathematics skills in spring, while mathematics interest in autumn was not a statistically significant predictor.

OLS Regression Results

Dep. Variable:	matisum_2	R-squared:	0.539
Model:	OLS	Adj. R-squared:	0.533
Method:	Least Squares	F-statistic:	89.86
Date:	Mon, 09 Nov 2026	Prob (F-statistic):	1.38e-26
Time:	16:06:53	Log-Likelihood:	-406.95
No. Observations:	157	AIC:	819.9
Df Residuals:	154	BIC:	829.1
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	6.6634	1.462	4.557	0.000	3.775	9.552
matisum_1	0.7004	0.054	12.879	0.000	0.593	0.808
miema	0.5954	0.304	1.957	0.052	-0.006	1.197

Omnibus:	2.451	Durbin-Watson:	2.121
Prob(Omnibus):	0.294	Jarque-Bera (JB):	2.164
Skew:	0.131	Prob(JB):	0.339
Kurtosis:	3.512	Cond. No.	92.4

Table 7. Multiple linear regression results predicting mathematics skills in spring (Python: statsmodels, OLS).

4. Confirmatory Factor Analysis

4.1 Research question and model

The research question for the CFA was: *Do three observed indicators of mathematics interest measure a latent mathematics-interest construct in autumn and spring?*

To address this question, a two-factor model was specified. The first latent factor represents mathematics interest in autumn (mat_interest1), measured by three observed indicators

(mie1c, mie2c, and mie3c). The second latent factor represents mathematics interest in spring (mat_interest2), measured by three observed indicators (mie1c_2, mie2c_2, and mie3c_2).

The model was specified in Mplus as follows:

```
mat_interest1 BY mie1c mie2c mie3c;

mat_interest2 BY mie1c_2 mie2c_2 mie3c_2;

mat_interest1 mat_interest2;

mie1c mie2c mie3c mie1c_2 mie2c_2 mie3c_2;

mie1c_2 WITH mie1c;
```

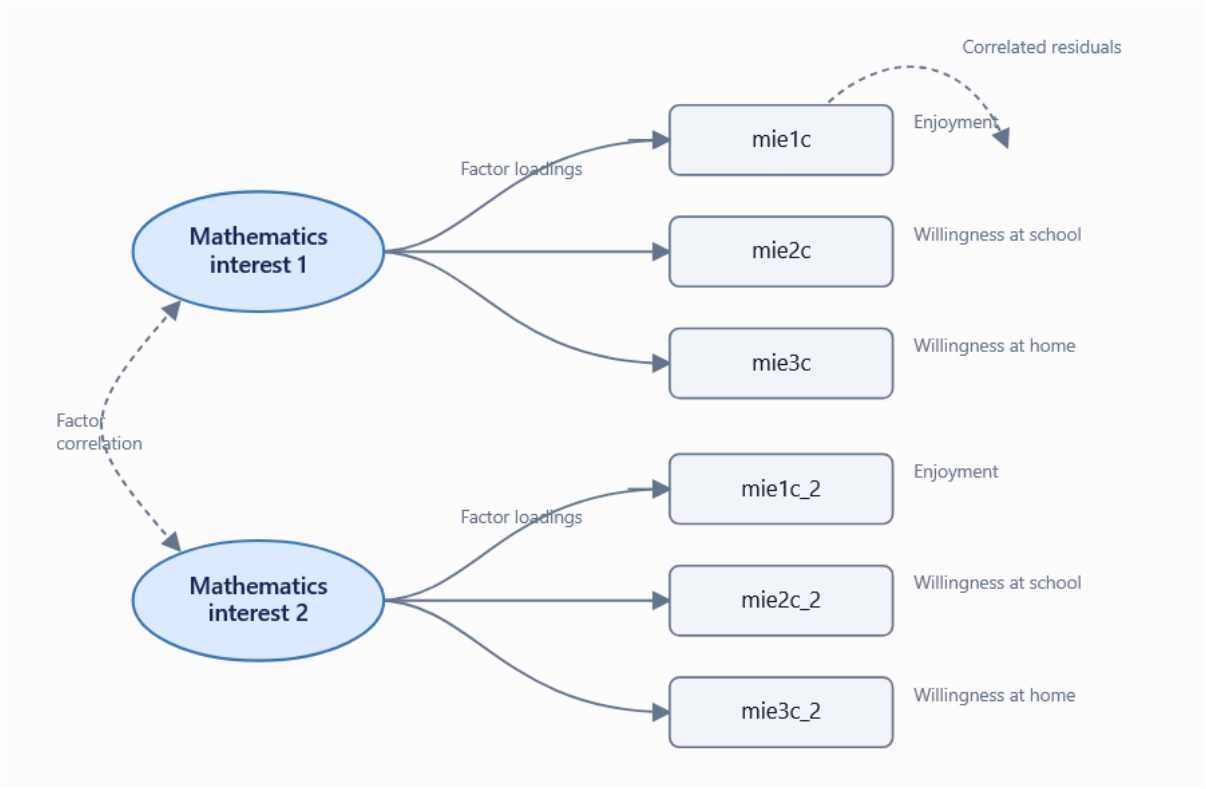


Figure 1. Path diagram of the two-factor confirmatory factor analysis model of mathematics interest in autumn and spring (ChatGPT).

The analysis was first conducted in Python using the semopy package. However, I faced difficulties with model estimation and missing-data handling in Python. Therefore, I also estimated the model in Mplus to compare the results and gain a better understanding of the model and its fit.

4.2 Model fit

The CFA model was first estimated in Python using the semopy package. Listwise deletion was used to handle missing data, resulting in an analytical sample of 156 participants out of the original 167. The model showed mixed fit. The CFI (0.962) and TLI (0.919) indicated reasonably

good fit, whereas the RMSEA (0.121) and chi-square test (p = 0.002) suggested that the model did not fit the data well.

Following these results, the same CFA model was estimated in Mplus to compare the results and investigate potential differences in estimation and missing-data handling. The Mplus analysis used the MLR estimator (Maximum Likelihood with robust standard errors). The results from Python and Mplus were broadly similar, although Mplus indicated somewhat better model fit. The Mplus results showed an RMSEA = 0.091 (90% CI [0.033, 0.148]), a chi-square p-value = 0.0203, a CFI = 0.958, and a TLI = 0.911. Additionally, the SRMR was 0.050, indicating good fit according to commonly used guidelines.

One possible explanation for the differences between the Python and Mplus results is their different approaches to handling missing data. Python used listwise deletion, whereas Mplus used FIML under the MLR estimator. FIML allows participants with partially missing observations to contribute available information to model estimation.

Overall, the model showed a reasonably acceptable, although not optimal, fit. The CFI, TLI, and SRMR indicated good fit, whereas the RMSEA remained relatively high and the chi-square test was statistically significant, suggesting some degree of model misfit. Nevertheless, considering the small size of sample and degrees of freedom, the model was used for the analysis. However, the fit indices indicate that the results should be interpreted with some caution.

	Value
DoF	7,000
DoF Baseline	15,000
chi2	22,891
chi2 p-value	0,002
chi2 Baseline	437,324
CFI	0,962
GFI	0,948
AGFI	0,888
NFI	0,948
TLI	0,919
RMSEA	0,121
AIC	27,707
BIC	70,405
LogLik	0,147

Table 8. Model fit indices for the two-factor confirmatory factor analysis model (Python: semopy).

```
Chi-Square Test of Model Fit

          Value                             16.581*
          Degrees of Freedom                    7
          P-Value                           0.0203
          Scaling Correction Factor         1.4221
            for MLR
```

Figure 2. Model fit indices for the two-factor confirmatory factor analysis model (Mplus).

```
RMSEA (Root Mean Square Error Of Approximation)

          Estimate                            0.091
          90 Percent C.I.                     0.033  0.148
          Probability RMSEA <= .05            0.104

CFI/TLI

          CFI                                 0.958
          TLI                                 0.911

Chi-Square Test of Model Fit for the Baseline Model

          Value                             245.553
          Degrees of Freedom                     15
          P-Value                           0.0000

SRMR (Standardized Root Mean Square Residual)

          Value                               0.050
```

Figure 2. Model fit indices for the two-factor confirmatory factor analysis model (Mplus).

4.3 Results

The parameter estimates showed that all six observed indicators loaded positively and significantly on their respective latent mathematics-interest factors ($p < 0.001$). The standardized factor loadings ranged from 0.517 to 0.779 for autumn and from 0.751 to 0.945 for spring. This indicates that all indicators contributed to measuring mathematics interest, although the enjoyment and willingness indicators at home and school did not contribute equally strongly.

The correlation between the autumn and spring latent factors was positive and statistically significant (standardized estimate = 0.417, $p < 0.001$). This suggests that children with higher mathematics interest in autumn also tended to report higher mathematics interest in spring.

Additionally, the residual covariance between the two enjoyment indicators (mie1c and mie1c_2) was negative and statistically significant (standardized estimate = −0.093, $p = 0.006$). This indicates a small negative association between the indicators beyond what is explained by the latent factors.

Overall, the results support the use of the six indicators to measure mathematics interest at both time points. However, the factor loadings varied in strength, particularly for the autumn indicators.

lval		rval	Estimate	Est. Std	Std. Err	z-value	p-value
mie1c	~	mat_interest1	1,000	0,774	-	-	-
mie2c	~	mat_interest1	0,931	0,779	0,144	6,489	0,000
mie3c	~	mat_interest1	0,698	0,517	0,128	5,456	0,000
mie1c_2	~	mat_interest2	1,000	0,945	-	-	-
mie2c_2	~	mat_interest2	0,847	0,866	0,059	14,390	0,000

mie3c_2	~	mat_interest2	0,749	0,751	0,065	11,605	0,000
mat_interest1	~~	mat_interest2	0,369	0,417	0,092	3,995	0,000
mat_interest1	~~	mat_interest1	0,707	1,000	0,155	4,563	0,000
mat_interest2	~~	mat_interest2	1,104	1,000	0,148	7,466	0,000
mie1c	~~	mie1c_2	-0,112	-0,093	0,041	-2,744	0,006
mie1c	~~	mie1c	0,475	0,402	0,110	4,315	0,000
mie1c_2	~~	mie1c_2	0,131	0,106	0,054	2,418	0,016
mie2c	~~	mie2c	0,399	0,394	0,092	4,321	0,000
mie2c_2	~~	mie2c_2	0,264	0,250	0,047	5,580	0,000
mie3c	~~	mie3c	0,948	0,733	0,119	7,960	0,000
mie3c_2	~~	mie3c_2	0,478	0,436	0,062	7,770	0,000

Table 9. Parameter estimates for the two-factor confiratory factor analysis model (Python: semopy).

STDYX Standardization

	Estimate	S.E.	Est./S.E.	Two-Tailed P-Value
MAT_INTE BY				
MIE1C	0.760	0.079	9.611	0.000
MIE2C	0.742	0.093	8.012	0.000
MIE3C	0.508	0.082	6.164	0.000
MAT_INTE BY				
MIE1C_2	0.942	0.031	30.606	0.000
MIE2C_2	0.866	0.052	16.787	0.000
MIE3C_2	0.759	0.063	12.042	0.000
MAT_INTE WITH				
MAT_INTERE	0.448	0.117	3.822	0.000
MIE1C_2 WITH				
MIE1C	-0.467	0.210	-2.225	0.026
Intercepts				
MIE1C	4.015	0.366	10.958	0.000
MIE2C	4.179	0.362	11.557	0.000
MIE3C	3.637	0.302	12.054	0.000
MIE1C_2	3.794	0.344	11.033	0.000
MIE2C_2	4.100	0.398	10.311	0.000
MIE3C_2	3.972	0.343	11.566	0.000
Variances				
MAT_INTERE	1.000	0.000	999.000	999.000
MAT_INTERE	1.000	0.000	999.000	999.000
Residual Variances				
MIE1C	0.423	0.120	3.521	0.000
MIE2C	0.449	0.138	3.262	0.001
MIE3C	0.742	0.084	8.851	0.000
MIE1C_2	0.113	0.058	1.943	0.052
MIE2C_2	0.251	0.089	2.806	0.005
MIE3C_2	0.423	0.096	4.419	0.000

Figure 4. Standardized parameter estimates for the two-factor confirmatory factor analysis model (Mplus).

5. Conclusion

The purpose of this assignment was to examine the relationship between children's mathematics interest and mathematics skills using descriptive statistics, regression analysis, and confirmatory factor analysis.

The descriptive statistics showed that children generally reported relatively high levels of mathematics interest in both autumn and spring. Mathematics skills also had higher mean and median scores in spring than in autumn, although this descriptive difference alone does not establish statistically significant improvement.

The regression analysis showed that mathematics skills in autumn were a strong and statistically significant predictor of mathematics skills in spring. However, mathematics interest in autumn was not a statistically significant predictor of mathematics skills in spring after controlling for skills in autumn. Therefore, the results did not provide sufficient evidence that mathematics interest independently predicts later mathematics skills.

The CFA results provided support for using the three observed indicators to measure mathematics interest at both measurement times. All six indicators had positive and statistically significant factor loadings, and the autumn and spring latent factors were positively correlated. However, the model fit was mixed, particularly because the RMSEA remained relatively high. Therefore, the CFA findings should be interpreted with some caution.

Overall, the findings suggest that children's earlier mathematics skills are more strongly associated with their later mathematics skills than their earlier mathematics interest is. The CFA results also indicate that mathematics interest can be represented as a latent construct at both time points, although the measurement model could potentially be improved. These findings should be interpreted in light of the sample size and the limitations of the analyses.